

Java Bank Application Problem Statements

Problem Statement 1: Bank Account Creation and Validation

Scenario:

A bank wants to automate customer account creation. Validate all details before creating the account. If any validation fails, throw an appropriate custom exception with a meaningful message.

Requirements:

Create a BankAccount class with fields:

- accountNumber
- customerName
- mobileNumber
- email
- initialDeposit
- accountType (Savings/Current)

Validation Rules:

1. Account number must contain exactly 10 digits.
2. Customer name should contain only alphabets and spaces.
3. Mobile number must contain exactly 10 digits and start with 6, 7, 8, or 9.
4. Email should contain '@' and end with '.com'.
5. Initial deposit: Savings $\geq$ ₹1000, Current $\geq$ ₹5000.
6. Account type must be Savings or Current.

Custom Exceptions:

- InvalidAccountNumberException
- InvalidCustomerNameException
- InvalidMobileNumberException
- InvalidEmailException
- InvalidDepositException
- InvalidAccountTypeException

Expected:

Display account details on success; otherwise display the custom exception message.

Problem Statement 2: Money Transfer System

Scenario:

Implement a money transfer module that validates business rules before processing a transaction.

Requirements:

Create a Transaction class with:

- senderAccountNumber
- receiverAccountNumber
- transferAmount
- transactionPin
- accountBalance

Validation Rules:

1. Sender and receiver account numbers must each contain exactly 10 digits.
2. Sender and receiver accounts cannot be the same.
3. Transfer amount must be greater than 0.
4. Maximum transfer amount is ₹50,000.
5. Sender must maintain a minimum balance of ₹1000 after transfer.
6. PIN must contain exactly 4 digits.

Custom Exceptions:

- InvalidSenderAccountException
- InvalidReceiverAccountException
- SameAccountTransferException
- InvalidTransferAmountException
- InsufficientBalanceException
- InvalidPinException

Expected:

Display successful transaction details or the appropriate custom exception message.